LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	A/c controls systems; mechanical, power assisted fully powered & fly-by-wire type of controls surfaces.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	A/c controls systems; mechanical, power assisted fully powered & fly-by-wire type of controls surfaces.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB schematics; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary flight controls?
(b) Define the function of the elevator.
(c) List two types of control systems used in modern aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Mechanical Control Systems • Cables and Push-pull Rods • Bell Cranks and Pulleys • Friction and Lubrication	L & Ds	PPT, IB	15 min
(b)	• Power Assisted Systems • Hydraulic Boost Mechanisms • Electro-Hydraulic Actuators • Fail-safe Features	L & Ds	PPT, PDF	20 min
(c)	• Fully Powered Systems • Fly-by-Wire Architecture • Flight Control Computers • Redundancy and Safety	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between mechanical and fly-by-wire controls.
(b) Describe the role of flight control computers in a fully powered system.
(c) Identify one safety feature common in power-assisted systems.

12. Home Assignment and Follow Up	Prepare a brief note comparing the mechanical linkage of LP 02 with the electronic signals used in Fly-by-Wire systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Control system components.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Engineer, P No. 12345	Senior Instructor, Chief Engineer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min